



Lab Sheet 4

Database Management System Lab (BTCS203P) 2026-2025)

Objective:

To understand and practice SQL Operators.

Prerequisites:

1. Basic knowledge of SQL.
2. MySQL installed and set up.

Lab Sheet 4: SQL Operators

```
CREATE DATABASE DBMS_Lab4;  
USE DBMS_Lab4;
```

-- Student table

```
CREATE TABLE Student (  
    Student_ID INT PRIMARY KEY,  
    Student_Name VARCHAR(50),  
    Course VARCHAR(50),  
    City VARCHAR(50),  
    Age INT,  
    Marks INT  
);
```

```
INSERT INTO Student VALUES  
(1, 'Aarav', 'Data Science', 'Delhi', 20, 85),  
(2, 'Priya', 'AI ML', 'Roorkee', 19, 92),  
(3, 'Rahul', 'Data Science', 'Delhi', 21, 76),  
(4, 'Ananya', 'Cyber Security', 'Haridwar', 20, 88),  
(5, 'Riya', 'Data Science', 'Dehradun', 19, 95),  
(6, 'Karan', 'AI ML', 'Delhi', 21, 67);
```

-- Employee table

```
CREATE TABLE Employee (  
    Employee_ID INT PRIMARY KEY,  
    Employee_Name VARCHAR(50),  
    Department VARCHAR(50),  
    Salary DECIMAL(10,2),  
    City VARCHAR(50),  
    Manager_Name VARCHAR(50)  
);
```

```
INSERT INTO Employee VALUES  
(101, 'Amit', 'IT', 45000, 'Delhi', 'Rajesh'),  
(102, 'Neha', 'HR', 38000, 'Roorkee', 'Suman'),  
(103, 'Rohan', 'Finance', 52000, 'Delhi', 'Rajesh'),  
(104, 'Sneha', 'IT', 60000, 'Haridwar', 'Vikas'),
```

```
(105, 'Pooja', 'HR', 42000, 'Dehradun', NULL),
(106, 'Vivek', 'Finance', 75000, 'Delhi', 'Rajesh');
SELECT * FROM Employee ;
```

-- Product table

```
CREATE TABLE Product (
    Product_ID INT PRIMARY KEY,
    Product_Name VARCHAR(50),
    Price DECIMAL(10,2),
    Quantity INT
);
```

```
INSERT INTO Product VALUES
```

```
(1, 'Laptop', 55000, 2),
(2, 'Mouse', 800, 5),
(3, 'Keyboard', 1500, 3),
(4, 'Monitor', 12000, 2),
(5, 'Headphones', 2500, 4);
SELECT * FROM Product;
```

STUDENT TABLE

Result Grid						
Filter Rows:						
Edit:						
	Student_ID	Student_Name	Course	City	Age	Marks
▶	1	Aarav	Data Science	Delhi	20	85
	2	Priya	AI ML	Roorkee	19	92
	3	Rahul	Data Science	Delhi	21	76
	4	Ananya	Cyber Security	Haridwar	20	88
	5	Riya	Data Science	Dehradun	19	95
	6	Karan	AI ML	Delhi	21	67
✱	NULL	NULL	NULL	NULL	NULL	NULL

EMPLOYEES TABLE

Result Grid						
Filter Rows:						
Edit:						
Export/Import:						
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	102	Neha	HR	38000.00	Roorkee	Suman
	103	Rohan	Finance	52000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
	105	Pooja	HR	42000.00	Dehradun	NULL
	106	Vivek	Finance	75000.00	Delhi	Rajesh
✱	NULL	NULL	NULL	NULL	NULL	NULL

PRODUCT TABLE

Result Grid  Filter Rows: Ec				
	Product_ID	Product_Name	Price	Quantity
▶	1	Laptop	55000.00	2
	2	Mouse	800.00	5
	3	Keyboard	1500.00	3
	4	Monitor	12000.00	2
	5	Headphones	2500.00	4
	NULL	NULL	NULL	NULL

Arithmetic Operators

1. Perform addition of two numbers using

SELECT 25 + 15 AS Addition;

Result Grid  Filter Rows:	
	Addition
▶	40

2. Perform subtraction using -.

SELECT 50 - 20 AS Subtraction;

Result Grid  Filter Rows:	
	Subtraction
▶	30

3. Perform multiplication using *.

SELECT 12 * 5 AS Multiplication;

Result Grid  Filter Rows:	
	Multiplication
▶	60

4. Perform division using /.

SELECT 100 / 5 AS Division;

Result Grid		Filter Rows:
	Division	
▶	20.0000	

5. Calculate the total amount as Price × Quantity.

```
SELECT
    Product_Name,
    Price,
    Quantity,
    Price * Quantity AS Total_Amount
FROM Product;
```

Result Grid



Filter Rows:

Export

	Product_Name	Price	Quantity	Total_Amount
▶	Laptop	55000.00	2	110000.00
	Mouse	800.00	5	4000.00
	Keyboard	1500.00	3	4500.00
	Monitor	12000.00	2	24000.00
	Headphones	2500.00	4	10000.00

Logical Operators

6. Demonstrate AND with two conditions.

```
SELECT *
FROM Employee
WHERE City = 'Delhi'
AND Department = 'IT';
```

Result Grid

Filter Rows:

Edit:

Export/Import:

	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
●	NULL	NULL	NULL	NULL	NULL	NULL

7. Demonstrate OR with two conditions.

```
SELECT *
FROM Employee
WHERE Department = 'IT'
OR Department = 'HR';
```

Result Grid						
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	102	Neha	HR	38000.00	Roorkee	Suman
	104	Sneha	IT	60000.00	Haridwar	Vikas
	105	Pooja	HR	42000.00	Dehradun	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL

8. Demonstrate NOT with a condition.

```
SELECT *
FROM Employee
WHERE NOT City = 'Delhi';
```

Result Grid						
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	102	Neha	HR	38000.00	Roorkee	Suman
	104	Sneha	IT	60000.00	Haridwar	Vikas
	105	Pooja	HR	42000.00	Dehradun	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL

9. Find students belonging to a specific course AND city.

```
SELECT *
FROM Student
WHERE Course = 'Data Science'
AND City = 'Delhi';
```

Result Grid						
	Student_ID	Student_Name	Course	City	Age	Marks
▶	1	Aarav	Data Science	Delhi	20	85
	3	Rahul	Data Science	Delhi	21	76
•	NULL	NULL	NULL	NULL	NULL	NULL

10. Find employees belonging to either of two departments.

```
SELECT *
FROM Employee
WHERE Department = 'IT'
OR Department = 'Finance';
```

Result Grid						
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	103	Rohan	Finance	52000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
	106	Vivek	Finance	75000.00	Delhi	Rajesh
•	NULL	NULL	NULL	NULL	NULL	NULL

Comparison Operators

11. Demonstrate the = operator.

```
SELECT *  
FROM Employee  
WHERE Department = 'IT';
```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
•	NULL	NULL	NULL	NULL	NULL	NULL

12. Demonstrate the > operator.

```
SELECT *  
FROM Student  
WHERE Marks > 80;
```

Result Grid						
		Filter Rows:		Edit:		Ex
	Student_ID	Student_Name	Course	City	Age	Marks
▶	1	Aarav	Data Science	Delhi	20	85
	2	Priya	AI ML	Roorkee	19	92
	4	Ananya	Cyber Security	Haridwar	20	88
	5	Riya	Data Science	Dehradun	19	95
•	NULL	NULL	NULL	NULL	NULL	NULL

13. Demonstrate the < operator.

```
SELECT *  
FROM Student  
WHERE Marks < 80;
```

Result Grid						
		Filter Rows:		Edit:		
	Student_ID	Student_Name	Course	City	Age	Marks
▶	3	Rahul	Data Science	Delhi	21	76
	6	Karan	AI ML	Delhi	21	67
•	NULL	NULL	NULL	NULL	NULL	NULL

14. Demonstrate >= and <=.

```
SELECT *  
FROM Student  
WHERE Marks >= 80;  
SELECT *  
FROM Student  
WHERE Marks <= 80;
```

Result Grid						
		Filter Rows:		Edit:		
	Student_ID	Student_Name	Course	City	Age	Marks
▶	3	Rahul	Data Science	Delhi	21	76
	6	Karan	AI ML	Delhi	21	67
•	NULL	NULL	NULL	NULL	NULL	NULL

15. Demonstrate the <>/!= operator.

```
SELECT *
FROM Employee
WHERE Department <> 'HR';
```

Result Grid						
		Filter Rows:		Edit:		
				Export/Import:		
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	103	Rohan	Finance	52000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
	106	Vivek	Finance	75000.00	Delhi	Rajesh
•	NULL	NULL	NULL	NULL	NULL	NULL

Special Operators

16. Demonstrate BETWEEN for salary range.

```
SELECT *
FROM Employee
WHERE Salary BETWEEN 40000 AND 60000;
```

Result Grid						
		Filter Rows:		Edit:		
				Export/Import:		
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	103	Rohan	Finance	52000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
	105	Pooja	HR	42000.00	Dehradun	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL

17. Demonstrate IN for selecting multiple cities.

```
SELECT *
FROM Employee
WHERE City IN ('Delhi', 'Roorkee', 'Haridwar');
```

Result Grid						
		Filter Rows:		Edit:		
				Export/Import:		
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	102	Neha	HR	38000.00	Roorkee	Suman
	103	Rohan	Finance	52000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
	106	Vivek	Finance	75000.00	Delhi	Rajesh
•	NULL	NULL	NULL	NULL	NULL	NULL

18. Demonstrate NOT IN.

```
SELECT *  
FROM Employee  
WHERE City NOT IN ('Delhi', 'Roorkee');
```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	104	Sneha	IT	60000.00	Haridwar	Vikas
	105	Pooja	HR	42000.00	Dehradun	NULL
•	NULL	NULL	NULL	NULL	NULL	NULL

19. Demonstrate LIKE for names beginning with a particular letter.

```
SELECT *  
FROM Student  
WHERE Student_Name LIKE 'A%';
```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
	Student_ID	Student_Name	Course	City	Age	Marks
▶	1	Aarav	Data Science	Delhi	20	85
	4	Ananya	Cyber Security	Haridwar	20	88
•	NULL	NULL	NULL	NULL	NULL	NULL

20. Demonstrate IS NULL and IS NOT NULL.

```
SELECT *  
FROM Employee  
WHERE Manager_Name IS NULL;
```

```
SELECT *  
FROM Employee  
WHERE Manager_Name IS NOT NULL;
```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
	Employee_ID	Employee_Name	Department	Salary	City	Manager_Name
▶	101	Amit	IT	45000.00	Delhi	Rajesh
	102	Neha	HR	38000.00	Roorkee	Suman
	103	Rohan	Finance	52000.00	Delhi	Rajesh
	104	Sneha	IT	60000.00	Haridwar	Vikas
	106	Vivek	Finance	75000.00	Delhi	Rajesh
•	NULL	NULL	NULL	NULL	NULL	NULL

Set Operations

21. Demonstrate UNION using two compatible queries.

```
SELECT City  
FROM Student
```

```
UNION
```

```
SELECT City  
FROM Employee;
```

Result Grid	
	City
▶	Delhi
	Roorkee
	Haridwar
	Dehradun

22. Demonstrate UNION ALL. 23. Demonstrate INTERSECT where supported by the DBMS.

```
SELECT City
FROM Student
```

UNION ALL

```
SELECT City
FROM Employee;
```

Result Grid	
	City
	Haridwar
	Dehradun
	Delhi
	Delhi
	Roorkee
	Delhi
	Haridwar
	Dehradun
	Delhi

23. . Demonstrate INTERSECT where supported by the DBMS.

```
SELECT DISTINCT s.City
FROM Student s
INNER JOIN Employee e
ON s.City = e.City;
```

Result Grid	
	City
▶	Delhi
	Roorkee
	Haridwar
	Dehradun

24. Demonstrate EXCEPT/MINUS where supported.

```
SELECT DISTINCT s.City
FROM Student s
```

```

WHERE NOT EXISTS (
    SELECT 1
    FROM Employee e
    WHERE e.City = s.City
);

```

25. Perform a suitable problem using multiple operators together, such as Arithmetic + Logical + Comparison + Special operators.

```

SELECT *
FROM Student
WHERE
    (Course = 'Data Science' OR Course = 'AI ML')
    AND Marks >= 80
    AND City IN ('Delhi', 'Roorkee')
    AND Marks BETWEEN 80 AND 100;

```

Result Grid				
Filter Rows:				
	Product_Name	Price	Quantity	Total_Amount
▶	Laptop	55000.00	2	110000.00
	Monitor	12000.00	2	24000.00
	Headphones	2500.00	4	10000.00

Submission:

Take screenshots of each step's output.
Save the SQL scripts in a text file.